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Applied AI researcher and engineer building production-grade AI systems across LLMs, retrieval, agentic workflows, and evaluation infrastructure: from research to reliable real-world deployment.

PROFESSIONAL EXPERIENCE

AI Lead

Oct. 2025 – Present

ErisAI

- Architect and develop **Synapse**, a research-driven AI infrastructure layer for multi-agent workflows, integrating hybrid retrieval, multi-turn memory, MCP-based tool orchestration, intelligent model routing, and evaluation and observability protocols using Ragas and LangFuse to catch traceability and knowledge-access failures. ([Synapse](#))
- Deployed **Synapse**, incorporating a LoRA fine-tuned model via Unsloth, across multiple enterprise AI applications including legal and investigative platforms, processing over 30M+ private documents with grounded retrieval and traceable responses. ([Use case](#))
- Built **Ownify AI**, a multi-tenant white-label platform on top of Synapse with structural tenant isolation, per-tenant billing and usage metering, and workspace branding for organizations running their own AI product. ([Product](#))

AI Researcher

Sep. 2024 – Oct. 2025

VOLV AI

- Researched and developed a 3D virtual try-on SDK using neural rendering, GANs, and diffusion models; the system was adopted by fashion brands and increased user engagement by 40%. ([Demo](#))
- Developed controllable 2D and 3D digital human systems with pose estimation and generative modeling, supporting personalized experiences and reducing product return rates by 30%.
- Prototyped multimodal LLM systems integrating language, vision, and audio for cross-modal reasoning and instruction-following in customer support and interactive applications.

Research AI Engineer

Jan. 2023 – Dec. 2023

BHuman AI

- Built scalable generative video pipelines using neural reenactment, facial motion transfer, and voice cloning, reducing production costs by 40% for a platform serving 90K+ users and enterprise media clients. ([Use cases](#))
- Researched and deployed audio-driven lip-sync models using GANs and super-resolution, improving visual fidelity and establishing a core product capability and primary revenue stream.
- Integrated LLMs with persona-driven avatars and developed retrieval-based conversational systems for enterprise clients, combining private knowledge, persona conditioning, and response evaluation for real-world deployments.

Intern Machine Learning Engineer

Aug. 2021 – Nov. 2021

NAECO Blue GmbH

[\[Profile\]](#)

- Developed machine learning models to improve the spatial and temporal resolution of weather data for solar and wind energy forecasting.
- Built a data pipeline that reduced R&D time by nearly 50% and supported forecasting workflows later adopted by 5+ European meteorological agencies.

SELECTED PROJECTS	Synapse (2026). Research-driven AI system integrating retrieval, memory, routing, evaluation, and runtime controls for reliable model behavior.
	Ownify AI (2026). Multi-tenant white-label AI platform built on Synapse, with structural tenant isolation, per-tenant billing, and operational tooling for organizations running their own AI product.
	PatchProof (2026). Framework for evidence-grounded patch verification using execution traces, behavioral validation, and correctness analysis beyond unit tests.
	DarwinPatch (2026). Verifier-grounded coding-agent repair framework combining bounded evidence, route-aware search, and regression gates, lifting solve rate from 0.6 to 0.9.
	GateGRPO (2026). Verifier-grounded reinforcement learning (GRPO) for code repair, replacing a learned reward model with the hard-gate verifier itself, beating SFT on held-out solve rate at lower search cost.
	WarmLead (2026). Cache-liveness scheduler for lead-plus-sidekick LLM agent harnesses, with a cost simulator and live API verification to reduce idle-period token spend.
	DGM-LLM (2025). Autonomous code self-improvement framework combining evolutionary search with LLM-guided mutation and multi-dimensional evaluation.
	EmbedVoiceLLM (2025). Efficient multimodal framework mapping speech representations directly into LLM embedding space for low-latency voice interaction.

SKILLS	• Programming: Python, C++, Bash, MATLAB • LLM Systems: RAG, Agentic Workflows, Custom Agent Orchestration, MCP (Model Context Protocol), Conversation Memory, vLLM, llama.cpp • Retrieval & Grounding: Hybrid Retrieval, Vector Search, Cross-Encoder Reranking, Hallucination Detection (NLI Verification), Structure-Aware Document Chunking, Zero-Shot Information Extraction, Ragas, LangFuse • Frameworks: LangChain, Haystack, FastAPI, Unsloth, TRL, OpenAI API, Anthropic API, Gemini API • Vector Databases: Qdrant, FAISS, pgvector • ML Fundamentals: PyTorch, Transformers, TensorFlow, NumPy, multimodal learning, LoRA, Quantization, SFT, Reinforcement Learning (RL, GRPO), model evaluation, experiment design • Infrastructure: Docker, Redis, Git, Linux, CI/CD (GitHub Actions), Systemd, Weights & Biases, Prometheus, Grafana, AWS, Azure, GCP • Computer Vision / 3D: OpenCV, PyTorch3D, MediaPipe, ONNX
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PUBLICATIONS	Taneem, U. J., & Ayesha, N. (2024). Beyond CNNs: Encoded Context for Image Inpainting with LSTMs and Pixel CNNs. International Journal of Innovations in Science & Technology, (IJIST) 6(5), 165–179, Special Issue ICTIS 2024.
	Taneem, U. J., & Zakira, I. (2022). HTML Code Generation from Images with Deep Neural Networks. Journal of Engineering and Applied Sciences (JEAS), UET Peshawar. (Young Undergraduate Researcher Award)

EDUCATION	University of Engineering and Technology Peshawar, Pakistan
	<i>Bachelor of Science in Computer Science, CGPA: 3.58/4.0</i> Sep. 2018 – Sep. 2022